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Inventor(s)

MAYHUGH; TERRY

SELF-CALIBRATING ANGLE SENSOR

Abstract

An integrated circuit (IC) includes first, second, and third magnetic sensors. An analog-to-digital converter (ADC) has an input coupled to the first, second, and third magnetic sensors, and has an output. A digital circuit has an input coupled to the output of the ADC. The digital circuit is configured to obtain first digital sensor values from the ADC, determine interim rotation matrices for rotation of the first digital sensor values, and determine a calibration matrix based on the interim rotation matrices.

Inventors: MAYHUGH; TERRY (RICHARDSON, TX)

Applicant: TEXAS INSTRUMENTS INCORPORATED (Dallas, TX)

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Background/Summary

BACKGROUND

[0001] Magnetic sensors can be used to calculate the angle of a rotation shaft, e.g., a rotating shaft

of a motor. A magnet positioned on the rotating shaft rotates with the shaft, and the magnetic sensors sense the rotating magnetic field of the magnet. The angle of the magnet relative to the magnetic sensors can be calculated and used as a feedback signal to a motor controller to control the angle of the motor's shaft. Misalignment between the magnetic sensors and the magnet may cause errors in the angle calculated from the sensed magnetic field.

SUMMARY

[0002] In one example, an integrated circuit (IC) includes first, second, and third magnetic sensors. An analog-to-digital converter (ADC) has an input coupled to the first, second, and third magnetic sensors, and has an output. A digital circuit has an input coupled to the output of the ADC. The digital circuit is configured to obtain first digital sensor values from the ADC, determine interim rotation matrices for rotation of the first digital sensor values, and determine a calibration matrix based on the interim rotation matrices.

[0003] In another example, an IC includes magnetic sensors and an ADC having an input coupled to the magnetic sensors. The ADC has an output. A digital circuit has an input coupled to the output of the ADC. The digital circuit is configured to apply a rotation correction to digital sensor values from the ADC to thereby determine rotation-corrected sensor values and then determine an angle based on the rotation-corrected sensor values.

[0004] In yet another example, a method includes obtaining values from magnetic sensors and determining interim rotation matrices for rotation of the sensor values. The method also includes determining a calibration matrix based on the interim rotation matrices.

[0005] In another example, a method includes applying a rotation correction to values from magnetic sensors to determine rotation-corrected sensor values. The method further includes determining an angle based on the rotation-corrected sensor values.

[0006] Another example is directed to a method that includes obtaining first values from magnetic sensors and determining interim rotation matrices for rotation of the first values. The method further includes determining a calibration matrix based on the interim rotation matrices, applying the calibration matrix to second values from the magnetic sensors to determine rotation-corrected sensor values, and determining an angle based on the rotation-corrected sensor values.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0007] FIG. 1 is a system diagram including a motor, a magnet, and a self-calibrating angle sensor device, in an example.

[0008] FIGS. 2A-2D are diagrams illustrating different types of misalignment errors between the magnet and self-calibrating angle sensor device.

[0009] FIG. 3 is a graph of a sensor constellation with misalignment errors.

[0010] FIG. 4 is a block diagram of the self-calibrating angle sensor device of FIG. 1, in an example.

[0011] FIG. 5 is a block diagram of a digital circuit usable to perform a calibration for the self-calibrating angle sensor device, in an example.

[0012] FIG. 6 is a block diagram of operations performed by the self-calibrating angle sensor device including a rotation correction, in an example.

[0013] FIG. 7 is a block diagram of the rotation correction of FIG. 6, in an example.

[0014] FIG. 8 is a block diagram of a linear transformation included in the rotation correction, in an example.

[0015] FIG. 9 is a block diagram of a quadratic transformation included in the rotation correction, in an example.

[0016] FIG. 10 is a flow diagram of how a calibration matrix is determined, in an example.

[0017] FIG. 11 is a flow diagram of to determine a three-dimensional rotation matrix, in an example.

[0018] FIG. 12 is a flow diagram of how to modify sensor values using the rotation matrix, in an example.

DETAILED DESCRIPTION

[0019] The same reference numbers or other reference designators are used in the drawings to designate the same or similar (either by function and/or structure) features.

[0020] FIG. 1 is a system diagram including a motor **110**, a magnet **120**, a self-calibrating angle sensor device **130**, a motor controller **140**, and an inverter **150**. In some examples, motor **110** may be a three-phase motor, and, under control by motor controller **140**, inverter **150** provides current to each phase of motor **110**. Motor **110** includes a rotating shaft **114** to which magnet **120** is attached. The diameter of magnet **120** may be exaggerated relative to the diameter of rotating shaft **114** for ease of view. In some examples the diameter of magnet **120** may match the diameter of rotating shaft **114**. Magnet **120** rotates as shaft **114** rotates. The rotating magnet also causes its magnetic field to rotate. Self-calibrating angle sensor device **130** includes multiple orthogonally-arranged magnetic sensors, e.g., Hall-effect sensors, to detect the rotating magnetic field. Self-calibrating angle sensor device **130** calculates the angle (Θ) of the magnetic field relative to the self-calibrating angle sensor device. Self-calibrating angle sensor device **130** may calculate the angle multiple times per revolution of shaft **114**. Motor control **140** obtains the angle Θ from self-calibrating angle sensor device **130** and uses the angle to determine when to commutate motor **110**. Inverter **150** may include a separate phase for each of the phases of motor **110**, and motor controller **140** changes the commutation of motor **110** by controlling the state of each of the inverter's phases.

[0021] Self-calibrating angle sensor device **130** may be positioned close to magnet **120** and centered with respect to magnet **120**. FIGS. 2A-2D are diagrams illustrating different types of misalignment errors that can occur between self-calibrating angle sensor device **130** and magnet **120**. Absent the self-calibrating benefit of self-calibrating angle sensor device **130**, misalignment errors can reduce the accuracy of the calculated angle Θ , and accordingly, detrimentally impact the ability of motor controller **140** to accurately control motor **110**. Self-calibrating angle sensor device **130** includes three orthogonally-arranged magnetic sensors. The three orthogonal magnetic sensors are arranged along X, Y, and Z axes within self-calibrating angle sensor device **130**. As described below, self-calibrating angle sensor device **130** uses the three magnetic sensors to calibrate itself to calculate angle Θ .

[0022] FIG. 2A is an example of an offset error. Self-calibrating angle sensor device **130** includes a center alignment point **135** to which the magnet **120** should be centered. In FIG. 2A, magnet **120** has a center point **137** about which the magnet rotates. In the offset error example of FIG. 2A, the magnet's center point **137** is not aligned to center alignment point **135** of self-calibrating angle sensor device **130**. The offset **D1** between the two points **135** and **137** causes an error in the calculation of angle Θ . FIG. 2B is an example of sensor tilt in which magnet **120** is not positioned parallel to self-calibrating angle sensor device **130**. FIG. 2C is an example of a run-out error in which the center of the rotating shaft **114** is centered on center alignment point **135** of self-calibrating angle sensor device **130** but the magnet itself is not centered on the rotating shaft **114**. As a result of the misalignment between magnet **120** and shaft **114**, magnet **120** rotates in an elliptical path about center alignment point **135**. FIG. 2D is an example of magnetic tilt in which magnet **120** is positioned at an angle with respect to rotating shaft **114**. Without the self-calibrating feature of self-calibrating angle sensor device **130**, any of the offset, sensor tilt, run-out, and magnetic tilt errors illustrated in FIGS. 2A-2D can cause a miscalculation of angle Θ . In some systems, multiple of the offset, sensor tilt, run-out, and magnetic tilt errors may be present.

[0023] The alignment errors may result from manufacturing tolerances that are intentionally included in the assembly process to mount magnet **120** on shaft **114** and self-calibrating angle sensor device **130** with respect to magnet **120**. In some cases, pick-and-place equipment is used to

place self-calibrating angle sensor device **130** on a printed circuit board (PCB). The accuracy of the positioning of self-calibrating angle sensor device **130** by the pick-and-place equipment is a function of the tolerances inherent in the equipment. Further, assembly of the die containing the magnetic sensors within the integrated circuit (IC) package can result in rotation, tilt, and/or offset errors.

[0024] Values obtained from the sensors **132**, **134**, and **136** at the same time are referred to herein as a coordinate. As shaft **114** turns, the self-calibrating angle sensor device **130** obtains multiple coordinates from its sensors. Multiple coordinates from the sensors **132**, **134**, and **136** are referred to as a constellation of coordinate (or constellation for simplicity).

[0025] Absent the misalignment errors of FIGS. 2A-2D, the magnetic field that rotates due to the rotating magnet will only have an amplitude in along two axes, for example, the axis of the X-axis sensor **132** and the axis of the Y-axis sensor **134**, and the Z-axis sensor **136** will not detect any magnetic field. Further, absent the misalignment errors, the constellation of X, Y coordinates will be a circle in the X, Y plane. FIG. 3 is a graph of a circle **310** on the X, Y plane. Each point on circle **310** represents a coordinate including a value from the X-axis sensor **132** and a value from the Y-axis sensor **134** obtained concurrently. An example point of such a coordinate is coordinate **314**, which includes X and Y sensor values **X1** and **Y1**. The angle of shaft **114** relative to the X-axis and Y-axis sensors **132** and **134** is shown as angle Θ in FIG. 3. Self-calibrating angle sensor device **130** may calculate angle Θ as:

$$[00001] \quad = \tan^{-1} \left(\frac{Y_1}{X_1} \right) \quad (\text{Eq. 1})$$

[0026] However, due to one or more misalignment errors, the rotating magnetic field will have a magnitude in all three axes. Further, the shape of the constellation of coordinates may be an ellipse, not a circle as in FIG. 3.

[0027] Self-calibrating angle sensor device **130** advantageously can calibrate itself given any of the possible offset, sensor tilt, run-out, and magnetic tilt alignment error that may be present. Without the ability for self-calibration, the system integrator of system **100** would be forced to perform a manual calibration of an angle sensor device using specialized equipment to accurately measure angle Θ and create a look-up table mapping accurate measurements of angle Θ to the angle calculated by the angle sensor device. The resulting look-up table would then be loaded into memory in the angle sensor device. The more data that is included in the look-up table, the more accurate would be the angle sensor device's ability to calculate angle Θ . Unfortunately, in addition to burdening the system integrator with this manual calibration process, large look-up tables require large amounts of memory in the angle sensor device thereby making the device large and expensive. Advantageously, the self-calibration capability of self-calibrating angle sensor device **130**, described herein, avoids the need for manual calibration and a large look-up table.

[0028] FIG. 4 is a block diagram of self-calibrating angle sensor device **130**, in an example. Self-calibrating angle sensor device **130** includes X-axis sensor **132**, Y-axis sensor **134**, Z-axis sensor **136**, a sample-and-hold and multiplexer circuit **402**, an amplifier and filter **408**, an analog-to-digital converter (ADC) **412**, a temperature sensor **416**, a digital circuit **420**, and an interface **424**. Sample-and-hold and multiplexer circuit **402** has inputs **402a**, **402b**, and **402c** coupled to respective X-axis sensor **132**, Y-axis sensor **134**, and Z-axis sensor **136**. Sample-and-hold and multiplexer circuit **402** has an output **402d** coupled to an input **408a** of amplifier and filter **408**. Amplifier and filter **408** has an output **408b** coupled to an input **412a** of ADC **412**. ADC **412** has an output **412b** coupled to an input **420a** of digital circuit **420**. Temperature sensor **416** is coupled to an input **420b** of digital circuit **420**. Digital circuit **420** has a terminal **420c** coupled to a terminal **424a** interface **424**. Interface **424** has a terminal **424b** to which motor controller **140** can be coupled.

[0029] Sample-and-hold and multiplexer circuit **402** may sample and hold any or all of the output signals from the X-axis sensor **132**, Y-axis sensor **134**, and Z-axis sensor **136**. In one example, sample-and hold and multiplexer circuit **402** may sample and hold the output signals from any pair

or all of the X-axis sensor **132**, Y-axis sensor **134**, and Z-axis sensor **136** approximately simultaneously. Sample-and-hold and multiplexer circuit **402** holds, e.g., charges a capacitor, the signal level of the sensors **132**, **134**, **136**. Each such held signal can then be sequentially processed through amplifier and filter **408**, ADC **412**, and digital circuit **420**. For example, the signal chain formed by amplifier and filter **408**, ADC **412**, and digital circuit **420** may process the held signal of one of the X-axis sensor **132**, Y-axis sensor **134**, or Z-axis sensor **136**, then the next held signal, and so forth. Amplifier and filter **408** amplifies and filters, e.g., low-pass filters, the signal. ADC **412** converts the amplified and filtered analog signal to a digital value for subsequent processing by digital circuit **420**. The example of FIG. 4 includes a single signal chain to sequentially process the sensor signals. Other examples include a separate a separate signal chain to process the three sensor signals concurrently.

[0030] Interface **424** provides an interface to which motor controller **140** can be coupled. Interface **424** may be any suitable type of interface such as a Single Edge Nibble Transmission (SENT) interface. Digital circuit **420** may calculate angle Θ , and motor controller **140** may retrieve the calculated angle Θ from self-calibrating angle sensor device **130** by way of interface **424**.

[0031] The components shown in the example of FIG. 4 may all be fabricated as part of the same semiconductor package. The components may all be fabricated on the same semiconductor die within the package. Alternatively, two or more of the components may be fabricated on separate dies, and the dies then packaged together as one device.

[0032] FIG. 5 is an example block diagram of digital circuit **420**. In this example, digital circuit **420** includes a processor **510** coupled to a memory device **514**. Memory device **514** is a non-transitory storage device such as random-access memory, read-only memory, etc. Machine instructions **518** are stored in memory device **514**. Processor **510** retrieves the machine instructions **518** from memory device **514** and executes the instructions. The functionality described herein as attributable to the digital circuit **420** can be performed by processor **510** through its execution of machine instructions **518**.

[0033] FIG. 6 is a flow diagram **600** illustrating a voltage offset correction **612**, a temperature calibration **616**, and a rotation correction **620** performed by digital circuit **420** on each coordinate of values from the X-axis sensor **132**, the Y-axis sensor **134**, and the Z-axis sensor **136**, for simplicity shown in FIG. 5 as the coordinate X, Y, and Z. In one example, the voltage offset correction **612**, temperature calibration **616**, and rotation correction **620** are performed by processor **510** executing machine instructions **518**. Each of the magnetic sensors may have a direct current (DC) voltage offset which detrimentally impacts the angle measurement. Voltage offset correction **612** receives the coordinate X, Y, and Z from the sensors (after being amplified, filtered, and digitized as described above) and determines the DC voltage offset for each sensor and subtracts the DC offset from the sensors' signals. The voltage offset-corrected output of offset correction **612** is the coordinate X', Y', and Z'. Temperature calibration **616** uses a reading from temperature sensor **416** to correct the X', Y', and Z' coordinate for changes in ambient temperature of self-calibrating angle sensor device **130** to produce the coordinate X'', Y'', and Z''.

[0034] Rotation correction **620** corrects the X'', Y'', and Z'' values due to any of the misalignment errors explained above. Rotation correction **620** rotates the constellation of coordinates from the X, Y, Z space to the X, Y plane, with little or no Z component, and also scales the resulting constellation in the X, Y plane to convert it from an ellipse to a circle. Rotation correction **620** applies a rotation correction to the coordinate X'', Y'', and Z''. In one example, the rotation correction includes a calibration matrix that is applied to the X'', Y'', and Z'' coordinate. The X'', Y'', and Z'' coordinate may be multiplied by the calibration matrix. The output of the rotation correction function **620** is X''' and Y'''. FIGS. 1-13G and the corresponding text below describe an example for how the calibration matrix can be determined.

[0035] In this example, it is assumed that self-calibrating angle sensor device **130** is oriented relative to magnetic **120** such that the X-axis sensor **132** and the Y-axis sensor **134** are orthogonal

to the rotating magnetic field and should be the primary sensors to detect the magnetic field, not the Z-axis sensor **136**. However, the calibration process described herein applies to whichever of the two orthogonal sensors are primarily orthogonal to the rotating magnetic field and rotates the constellation to the plane defined by those sensors.

[0036] The output X''' , Y''' coordinate from the rotation correction **620** is provided to coordinate rotation digital computer (CORDIC) **624** calculates angle Θ based on X''' and Y''' . In one example, CORDIC **624** may calculate angle Θ using Eq. (1) above. CORDIC **624** may be performed by processor **510** executing machine instructions **518**.

[0037] FIG. 7 is a block diagram of rotation correction **620**. Rotation correction **620** includes a linear transformation **708**, a quadratic transformation **715**, and adders **720** and **728**. The linear transformation converts the constellation of sensor coordinates from the X, Y, Z space to the X, Y plane with little or no Z component remaining and scales the resulting constellation to make it a circle. As described above, two or more of the misalignment errors of FIGS. 2A-2D may be present at the same time in any given system. If such multi-alignment errors are present, the resulting constellation resulting from the linear transformation **708** may still have a Z component. The quadratic formation **715** may be included to further reduce the Z component to further improve the accuracy of the calculated angle Θ .

[0038] The inputs to the linear transformation **708** are the X'' , Y'' , and Z'' coordinates from the temperature calibration **616**. The outputs of the linear transformation **708** are X_LT and Y_LT . The X_LT and Y_LT output coordinate from the linear transformation **708** is provided as input into the quadratic transformation **715**, the output of which is X_QT and Y_QT . Adder **720** adds X_LT and X_Q to produce X''' . Similarly, adder **728** adds Y_LT and Y_QT to produce Y''' . In an example in which the quadratic transformation **715** is not included or not performed, the output coordinate X_LT , Y_LT from the linear transformation **708** is the X''' , Y''' output of the rotation correction **620**—adders **720** and **728** are not included or not used in this example.

[0039] FIG. 8 is a block diagram of the linear transformation **708**. In this example, linear transformation **708** includes multipliers **802**, **804**, **806**, **808**, **810**, and **812** and adders **814** and **816**. Multiplier **802** multiplies X'' by a value $L_{sub.XX}$. The output value from multiplier **802** is provided to an input of adder **814**. Multiplier **804** multiplies Y'' by a value $L_{sub.XY}$. The output value from multiplier **804** is provided to another input of adder **814**. Multiplier **806** multiplies Z'' by a value $L_{sub.XZ}$. The output value from multiplier **806** is provided to yet another input of adder **814**. Adder **814** adds the values from multipliers **802**, **804**, and **806** to produce output value X_LT . Multiplier **808** multiplies X'' by a value $L_{sub.YX}$. The output value from multiplier **808** is provided to an input of adder **816**. Multiplier **810** multiplies Y'' by a value $L_{sub.YY}$. The output value from multiplier **810** is provided to another input of adder **816**. Multiplier **812** multiplies Z'' by a value $L_{sub.YZ}$. The output value from multiplier **812** is provided to yet another input of adder **816**. Adder **816** adds the values from multipliers **808**, **810**, and **812** to produce output value Y_LT .

[0040] The calibration matrix described above includes the values $L_{sub.XX}$, $L_{sub.XY}$, $L_{sub.XZ}$, $L_{sub.YX}$, $L_{sub.YY}$, and $L_{sub.YZ}$ arranged as a 3×2 matrix, which is multiplied, per linear transformation **708**, with a 1×3 vector including the coordinate X'' , Y'' , Z'' per Eq. (2) below.

$$[00002] \begin{bmatrix} X_LT \\ Y_LT \end{bmatrix} = \begin{bmatrix} X'' \\ Y'' \\ Z'' \end{bmatrix} * \begin{bmatrix} L_{XX} & L_{XY} & L_{XZ} \\ L_{YX} & L_{YY} & L_{YZ} \end{bmatrix} \quad (\text{Eq. 2})$$

[0041] FIG. 9 is a block diagram of the quadratic transformation **715**. In this example, the quadratic transformation includes multipliers **902**, **904**, **906**, **908**, **910**, **912**, **914**, **916**, and **918** and adders **920** and **922**. Multiplier **902** multiplies X'' by X'' , in other words, computes the square of X'' , to produce output XX . Similarly, multiplier **904** multiplies Y'' by Y'' to compute the square of Y'' as output YY . Multiplier **906** multiplies X'' by Y'' to compute XY .

[0042] Multiplier **908** multiplies XX by a value $Q_{sub.XXX}$ and provides the product to an input of

adder **920**. Multiplier **912** multiplies YY by a value Q.sub.XYY and provides the product to another input of adder **920**. Multiplier **916** multiplies XY by a value Q.sub.XXY and provides the product to another input of adder **920**. Adder **920** adds the products from multipliers **908**, **912**, and **916** to produce output X'''.

[0043] Multiplier **910** multiplies XX by a value Q.sub.YXX and provides the product to an input of adder **922**. Multiplier **914** multiplies YY by a value Q.sub.YYY and provides the product to another input of adder **922**. Multiplier **918** multiplies XY by a value Q.sub.YXY and provides the product to another input of adder **922**. Adder **922** adds the products from multipliers **910**, **914**, and **918** to produce output Y'''. In one example, the values of Q.sub.XXX, Q.sub.YXX, Q.sub.XYY, Q.sub.YYY, Q.sub.XXY, and Q.sub.YXY are determined by a least mean squares (LMS) technique to further reduce the Z component.

[0044] FIG. **10** is a flow diagram **1000** of an example process for determining the calibration matrix, which is the 3×2 matrix in Eq. (2) above. The calibration matrix is the combination, e.g., product, of multiple interim rotation matrices that implements the linear transformation **708**. Each interim rotation matrix is a rotation matrix that rotates the constellation of coordinates about one of the X, Y, or Z axes. The rotation matrices for rotating an object about any of the X, Y, or Z axes are:

$$[00003] R_x(\varphi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos(\varphi) & -\sin(\varphi) \\ 0 & \sin(\varphi) & \cos(\varphi) \end{bmatrix} \quad (\text{Eq. 3}) \quad R_y(\varphi) = \begin{bmatrix} \cos(\varphi) & 0 & \sin(\varphi) \\ 0 & 1 & 0 \\ -\sin(\varphi) & 0 & \cos(\varphi) \end{bmatrix} \quad (\text{Eq. 4})$$

$$R_z(\varphi) = \begin{bmatrix} \cos(\varphi) & -\sin(\varphi) & 0 \\ \sin(\varphi) & \cos(\varphi) & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (\text{Eq. 5})$$

where R.sub.x(φ), R.sub.y(φ), and R.sub.z(φ) are the rotation matrices about the X, Y, and Z axes, respectively, and angle φ is the angle through which the object is rotated about the corresponding axis. For example,) R.sub.x(30.sup.0) is a matrix used to rotate an object by 30 degrees about the X-axis. Rotation of an object through a particular angle φ about one of the axes includes substituting φ into the cos and sin operators in the corresponding rotation matrix and then multiplying the rotation matrix by any given coordinate on the object to be rotated and repeating the multiplication for any other desired coordinates on the object to be rotated.

[0045] FIG. **10** is now described in conjunction with FIGS. **13A-13G**. The operations of FIG. **10** may be performed by digital circuit **420**. FIGS. **13A-13C** graphically illustrate the rotation of the constellation of coordinates. Operation **1002** includes obtaining values from the magnetic sensors **132**, **134**, and **136**. The collection of values from all three magnetic sensors represents a coordinate, and multiple coordinates may be obtained in operation **1002**. As the constellation may be an ellipse in three-dimensional space, the number of coordinates obtained in operation **1002** may include 10 coordinates. With 10 coordinates, digital circuit **420** can determine the equation for the ellipse in three-dimensional space using the equation:

$$[00004] Ax^2 + Bxy + Cxz + Dy^2 + Eyz + Fz^2 + Gx + Hy + Iz + J = 0 \quad (\text{Eq. 6})$$

where A, B, C, D, E, F, G, H, I, and J are coefficients. In one example, digital circuit **420** determines the coefficients A through J based on 10 coordinates obtained of the ellipse.

[0046] In operation **1004**, digital circuit **420** determines the equation of the line normal to the plane of the ellipse formed by the 10 coordinates obtained in operation **1002**. In another example, because a plane can be specified based on three points, in operation **1002**, digital circuit **420** may obtain three coordinates and, then use those three coordinates to determine the equation for the plane in which the ellipse resides. The equation for the ellipse's plane can then be used by digital circuit **420** to determine the line normal to that plane.

[0047] FIG. **13A** shows an example of an ellipse **1310** formed by a constellation of the coordinates

obtained in operation **1002**. The units on the X, Y, and Z axes are milli-Teslas (mT). Although only a few (e.g., 3 or 10) coordinates are obtained in operation **1002**, the ellipse **1310** on which those coordinates reside is shown in FIG. **13A**. Line **1315** is the line normal to the plane of ellipse **1310**, and is referred to as normal line **1315**. FIG. **13A** also identifies points **1321** and **1322** on the constellation. Point **1321** corresponds to a “zero-degree” point, and point **1322** corresponds to a 90 degree point relative to point **1321**. The zero-degree point **1321** may represent point on the X-axis (0 degrees) but may not exactly align to the pole (e.g. north pole) of magnet **120** during the installation of the self-calibrating angle sensor device **130** relative to magnet **120**.

[0048] In operation **1006**, digital circuit **420** rotates the constellation about the Z-axis to align the ellipse so that normal line **1315** is in the X, Z plane as shown in FIG. **13B**. Digital circuit **420** may perform this rotation by determining the angle ϕ between line **1315** and the X-axis, computing the sin and cos values in the $R_{\text{sub.x}}(\phi)$ rotation matrix of Eq. (3) using the computed angle ϕ , and then multiplying rotation matrix $R_{\text{sub.x}}(\phi)$ by each of the coordinates obtained in operation **1002**. The rotation matrix $R_{\text{sub.x}}(\phi)$ determined using the computed angle ϕ is referred to as an interim rotation matrix (e.g., a first interim rotation matrix). Digital circuit **420** also saves the first interim matrix, e.g. in memory **514**.

[0049] FIG. **11** shows an example flow diagram **1100** for how a three-dimensional rotation matrix can be determined. The logic flow of FIG. **11** can be performed by digital circuit **420**. At operation **1102**, digital circuit **420** determines angle ϕ , as described above. At operation **1104**, digital circuit **420** then uses the angle ϕ to generate the cos and sin values in the corresponding rotation matrix, which digital circuit **420** then saves at operation **1106**.

[0050] In operation **1008**, digital circuit **420** rotates the constellation resulting from operation **1006** about the Y-axis to align the constellation normal to the Z-axis, as shown in FIG. **13C**. Digital circuit **420** may perform this rotation by determining a rotation matrix as shown in FIG. **11** using rotation matrix $R_{\text{sub.y}}(\phi)$ and determining the angle ϕ between the normal line **1315** of ellipse **1310** and the Y-axis.

[0051] Digital circuit **420** may then multiply the rotation matrix $R_{\text{sub.y}}(\phi)$ by each of the coordinates resulting from operation **1006**. The rotation matrix $R_{\text{sub.y}}(\phi)$ determined using the newly computed angle ϕ is referred to as an interim rotation matrix (e.g., a second interim rotation matrix). Digital circuit **420** also saves the second interim matrix, e.g., in memory **514**.

[0052] After the rotations of operations **1006** and **1008**, as shown in FIG. **13C** the ellipse **1310** is predominantly in just the X, Y plane and most or all of the Z component of the original ellipse **1310** in FIG. **13A** has been removed. Any remaining Z component in ellipse **1310** can be further reduced, if desired, by the digital circuit **420** performing the quadratic transformation **715**.

[0053] FIG. **13D** is a graph in just the X and Y axes of the ellipse **1310** from FIG. **13C**. At operation **1010**, digital circuit **420** determines the major and minor axes of ellipse **1310**. An ellipse's major axis is along its longest radius, and its minor axis is along its shortest radius. At operation **1012**, digital circuit **420** rotates the constellation of ellipse **1310** to align its major axis with the X- axis, as shown in FIG. **13E**. Digital circuit **420** may perform this two-dimensional rotation by rotating the ellipse **1310** about the Z-axis. A Z-axis rotation may be performed by multiplying the $R_{\text{sub.z}}(\phi)$ rotation matrix of Eq. (5) above by the points on ellipse **1310** while setting the value of Z in the result to a value of 0. The value of ϕ is the angle through which ellipse **1310** is to be rotated about the Z-axis. Digital circuit **420** saves this two-dimensional interim rotation matrix.

[0054] In operation **1014**, digital circuit **420** scales ellipse **1310** resulting from operation **1012** to convert it to a circle, as shown in FIG. **13F**. Digital circuit **420** may perform this operation by multiplying each coordinate by an interim scaling matrix which includes multiplying the Y-component of all (X,Y) points by A, where $A = \text{major axis}/\text{minor axis}$. Digital circuit **420** saves the interim scaling matrix, e.g., in memory **514**.

[0055] In operation **1016**, digital circuit **420** rotates ellipse **1310** resulting from operation **1014** to

align the zero-degree point **1321** with the X-axis, as shown in FIG. **13G**. Digital circuit **420** uses another instance of a two-dimensional interim rotation matrix to perform this rotation and saves the interim rotation matrix, e.g., in memory **514**.

[0056] At operation **1018**, digital circuit **420** determines the calibration matrix (3×2 matrix in Eq. (2)) based on the interim matrices determined in operations **1006**, **1008**, **1012**, **1014**, and **1016**. In one example, digital circuit determines the calibration matrices by multiplying the interim matrices together in the order they were generated in FIG. **10**.

[0057] The constellation resulting from the linear transformation of the operations illustrated in FIG. **10** may still have a Z component, albeit likely fairly small. The shape of the constellation with the Z component may have a “saddle” shape. If desired, to further reduce any remaining Z-component in the constellation, quadratic transformation **715** can be applied. An LMS technique can be used to determine the coefficients $Q_{\text{sub.XXX}}$, $Q_{\text{sub.YXX}}$, $Q_{\text{sub.XYY}}$, $Q_{\text{sub.YYY}}$, $Q_{\text{sub.XXY}}$, and $Q_{\text{sub.YXY}}$, described above, that sufficiently maps the saddle-shaped constellation to a planar circle thereby reducing any Z-component remaining from the linear transformation.

[0058] FIG. **12** is a flow diagram **1200** illustrating an example of how to use the calibration matrix generated according to the example of FIG. **11**. At operation **1202**, digital circuit **420** obtains values from the magnetic sensors **132**, **134**, and **136**. At operation **1204**, digital circuit **420** applies a rotation correction to the values from the magnetic sensors to determine rotation-corrected values. Digital circuit **420** may perform the rotation correction by applying the calibration matrix to the sensors. In a specific example, digital circuit **420** applies the rotation correction by multiplying the values by the calibration matrix. At operation **1206**, CORDIC **624**, which may be implemented by digital circuit **420**, determines the shaft angle Θ based on the rotation-corrected values, e.g., using Eq. (1). The offset correction **612** and temperature calibration **616** also may be performed by digital circuit **420**, e.g., between operations **1202** and **1204**.

[0059] In this description, the term “couple” may cover connections, communications, or signal paths that enable a functional relationship consistent with this description. For example, if device A generates a signal to control device B to perform an action: (a) in a first example, device A is coupled to device B by direct connection; or (b) in a second example, device A is coupled to device B through intervening component C if intervening component C does not alter the functional relationship between device A and device B, such that device B is controlled by device A via the control signal generated by device A.

[0060] Also, in this description, the recitation “based on” means “based at least in part on.” Therefore, if X is based on Y, then X may be a function of Y and any number of other factors.

[0061] A device that is “configured to” perform a task or function may be configured (e.g., programmed and/or hardwired) at a time of manufacturing by a manufacturer to perform the function and/or may be configurable (or reconfigurable) by a user after manufacturing to perform the function and/or other additional or alternative functions. The configuring may be through firmware and/or software programming of the device, through a construction and/or layout of hardware components and interconnections of the device, or a combination thereof.

[0062] As used herein, the terms “terminal”, “node”, “interconnection”, “pin” and “lead” are used interchangeably. Unless specifically stated to the contrary, these terms are generally used to mean an interconnection between or a terminus of a device element, a circuit element, an integrated circuit, a device or other electronics or semiconductor component.

[0063] A circuit or device that is described herein as including certain components may instead be adapted to be coupled to those components to form the described circuitry or device. For example, a structure described as including one or more semiconductor elements (such as transistors), one or more passive elements (such as resistors, capacitors, and/or inductors), and/or one or more sources (such as voltage and/or current sources) may instead include only the semiconductor elements within a single physical device (e.g., a semiconductor die and/or integrated circuit (IC) package)

and may be adapted to be coupled to at least some of the passive elements and/or the sources to form the described structure either at a time of manufacture or after a time of manufacture, for example, by an end-user and/or a third-party.

[0064] While certain elements of the described examples are included in an integrated circuit and other elements are external to the integrated circuit, in other example embodiments, additional or fewer features may be incorporated into the integrated circuit. In addition, some or all of the features illustrated as being external to the integrated circuit may be included in the integrated circuit and/or some features illustrated as being internal to the integrated circuit may be incorporated outside of the integrated. As used herein, the term “integrated circuit” means one or more circuits that are: (i) incorporated in/over a semiconductor substrate; (ii) incorporated in a single semiconductor package; (iii) incorporated into the same module; and/or (iv) incorporated in/on the same printed circuit board.

[0065] Uses of the phrase “ground” in the foregoing description include a chassis ground, an Earth ground, a floating ground, a virtual ground, a digital ground, a common ground, and/or any other form of ground connection applicable to, or suitable for, the teachings of this description. In this description, unless otherwise stated, “about,” “approximately” or “substantially” preceding a parameter means being within ± 10 percent of that parameter or, if the parameter is zero, a reasonable range of values around zero.

[0066] Modifications are possible in the described examples, and other examples are possible, within the scope of the claims.

Claims

1. An integrated circuit (IC), comprising: a first magnetic sensor; a second magnetic sensor; a third magnetic sensor; an analog-to-digital converter (ADC) having an input coupled to the first, second, and third magnetic sensors, and having an output; and a digital circuit having an input coupled to the output of the ADC, the digital circuit configured to: obtain first digital sensor values from the ADC; determine interim rotation matrices for rotation of the first digital sensor values; and determine a calibration matrix based on the interim rotation matrices.
2. The IC of claim 1, wherein the digital circuit is configured to determine the calibration matrix by multiplying together the interim rotation matrices.
3. The IC of claim 1, wherein the digital circuit is configured to determine the interim rotation matrices by: determining a direction normal to a constellation of the digital sensor values; determining a first interim rotation matrix to rotate the constellation about a first axis to produce a first rotated constellation; and determining a second interim rotation matrix to rotate the first rotated constellation about a second axis to produce a second rotated constellation.
4. The IC of claim 3, wherein the digital circuit is configured to determine the calibration matrix by multiplying the first interim rotation matrix by the second interim rotation matrix.
5. The IC of claim 3, wherein the digital circuit is configured to: determine a third interim rotation matrix to rotate the second rotated constellation about one of the first and second axes to produce a third rotated constellation; determine a scaling matrix to scale the third rotated constellation; and determine the calibration matrix based on the first interim rotation matrix, the second interim rotation matrix, the third interim rotation matrix, and the scaling matrix.
6. The IC of claim 5, wherein the digital circuit is configured to determine the calibration matrix by multiplying together the first interim rotation matrix, the second interim rotation matrix, the third interim rotation matrix, and the scaling matrix.
7. The IC of claim 1, wherein the digital circuit is configured to correct second digital sensor values using the calibration matrix.
8. The IC of claim 7, wherein the digital circuit is configured to correct the second digital sensors values by multiplying the second digital sensor values by the calibration matrix.

- 9.** The IC of claim 7, wherein the digital circuit is configured to perform a least mean square (LMS) operation on a result of multiplying the second digital sensor values by the calibration matrix to map a planar circle to the result.
- 10.** An integrated circuit (IC), comprising: magnetic sensors; an analog-to-digital converter (ADC) having an input coupled to the magnetic sensors, and having an output; and a digital circuit having an input coupled to the output of the ADC, the digital circuit configured to: apply a rotation correction to digital sensor values from the ADC to determine rotation-corrected sensor values; and determine an angle based on the rotation-corrected sensor values.
- 11.** The IC of claim 10, wherein the digital circuit is configured to apply the rotation correction to the digital sensor values by applying a matrix to the digital sensor values.
- 12.** The IC of claim 10, wherein the digital circuit is configured to apply the rotation correction to the digital sensor values by multiplying the digital sensor values by a correction matrix.
- 13.** The IC of claim 10, wherein the digital circuit is configured to determine the rotation correction by: obtaining first digital sensor values from the ADC; determining interim rotation matrices for rotation of the first digital sensor values; and determining a calibration matrix based on the interim rotation matrices.
- 14.** The IC of claim 13, wherein the digital circuit is configured to determine the calibration matrix by multiplying the interim rotation matrices together to compute the calibration matrix.
- 15.** A method, comprising: obtaining values from magnetic sensors; determining interim rotation matrices for rotation of the sensor values; and determining a calibration matrix based on the interim rotation matrices.
- 16.** The method of claim 15, wherein obtaining values from the magnetic sensors includes obtaining values from first, second, and third orthogonally-arranged magnetic sensors.
- 17.** The method of claim 15, wherein determining the calibration matrix includes multiplying together the interim rotation matrices.
- 18.** A method, comprising: applying a rotation correction to values from magnetic sensors to determine rotation-corrected sensor values; and determining an angle based on the rotation-corrected sensor values.
- 19.** The method of claim 18, wherein applying the rotation correction to the values includes applying a matrix to the values.
- 20.** The method of claim 18, wherein applying the rotation correction to the values includes multiplying the values by a calibration matrix.
- 21.** A method, comprising: obtaining first values from magnetic sensors; determining interim rotation matrices for rotation of the first values; determining a calibration matrix based on the interim rotation matrices; applying the calibration matrix to second values from the magnetic sensors to determine rotation-corrected sensor values; and determining an angle based on the rotation-corrected sensor values.
- 22.** The method of claim 21, wherein the first values include values from a first, second, and third magnetic sensor, and the second values includes values from two magnetic sensors of the first, second, and third magnetic sensors.
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